

# FINAL EXPLANATION: CHEMISTRY GRADE 10

## *Student Assessment Document*

### FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 1.1 INTRODUCTION TO CHEMISTRY: FINAL EXPLANATION

#### Student Assessment Document

|              |       |
|--------------|-------|
| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

Think back to the chapati dough on the hot jiko in Lesson 1. In this Final Explanation you will use evidence from all ten lessons to answer our Driving Question: "How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?" Write your answers in the space under each prompt. Use full sentences, name specific Kenyan examples from our lessons, and give reasons for your ideas. Where a prompt asks for a claim, evidence and reasoning, make sure you include all three parts. Take your time: there is no single word answer, we are marking how well you explain.

#### Part 1: What happened to the chapati dough?

In Lesson 1 you watched raw chapati dough change on a hot jiko into a firm, golden, aromatic chapati, and you could not turn it back into dough. (a) Define Chemistry and matter in your own words. (b) Explain why we say the change to the chapati is irreversible, and why Chemistry is a better science than a simple physical change idea for explaining it. (c) Compare it with the water particle changes you saw in the PhET States of Matter simulation. How is that change different?

*Write your answer here:*

## Part 2: Branches of Chemistry and the Maillard reaction

**In Lesson 2 you studied seven Branch Cards. (a) Name the branch of Chemistry that best explains the browning of the chapati and say why. (b) Describe in simple terms what the Maillard reaction is and what it produced in the chapati. (c) Choose TWO other branches and give a Kenyan example for each of how that branch is used.**

*Write your answer here:*

### Part 3: Chemistry in agriculture, food, medicine and industry

Chemistry is at work long before and long after the chapati is cooked. (a) Using the ingredients of the chapati (wheat flour, water, oil, salt), explain TWO ways Chemistry is involved in Kenya's food chain, using examples from Lessons 3 and 4. (b) Choose one example from Lessons 4 or 5 (medicine, manufacturing, sport, energy, entertainment or the environment) and explain how it is similar to the chemical change in the chapati.

*Write your answer here:*

### Part 4: Drugs, substance use and tobacco – a claim, evidence and reasoning

A friend says: "Shisha and tobacco are natural plants, so they are safe." Write a Claim–Evidence–Reasoning (CER) response. (a) State your claim. (b) Give evidence about tar, nicotine and carbon monoxide from Lesson 8. (c) Use your Chemistry knowledge to explain your reasoning, and link it to the chapati phenomenon. (d) Using Lesson 7, explain what a drug, a prescription and a dosage are, and why dosage matters.

*Write your answer here:*

### Part 5: Answering the Driving Question

Now answer the full Driving Question: "How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?" Use at least **THREE** lessons as evidence. Include (a) how Chemistry explains the chapati, (b) how Chemistry connects to at least two areas of life (such as food, health, industry, careers or consumer protection), and (c) one example of a safer, smarter choice that a Kenyan could make because of Chemistry. Finally, say how your thinking has changed since your first prediction in Lesson 1.

Write your answer here:

| RUBRIC                                        |                                                                                                                                                                                           |                                                                                                                                           |                                                                                                              |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Criterion                                     | Excellent (4)                                                                                                                                                                             | Proficient (3)                                                                                                                            | Developing (1–2)                                                                                             |
| Use of the anchoring phenomenon               | Clearly explains the chapati change as an irreversible chemical change with new substances formed (including the Maillard reaction) and links it accurately to other examples throughout. | Identifies the chapati change as irreversible and chemical, with some explanation of why, and makes at least one link to another example. | Describes the chapati changing but does not explain why it is irreversible or does not link it to Chemistry. |
| Accurate Chemistry ideas and vocabulary       | Uses terms such as matter, chemical change, branches of Chemistry, drug, prescription and dosage correctly and precisely.                                                                 | Uses most terms correctly with minor gaps or imprecision.                                                                                 | Uses few key terms, or uses them incorrectly.                                                                |
| Evidence from the lessons and Kenyan contexts | Supports ideas with specific evidence from at least three lessons (e.g. Rift Valley fertilisers, KEBS, tobacco chemicals, Panadol and Metformin labels).                                  | Gives some specific evidence from two or more lessons.                                                                                    | Gives general statements with little or no specific evidence.                                                |
| Reasoning and CER structure                   | Presents a clear claim supported by evidence and logical reasoning that explains why the evidence supports the claim.                                                                     | Gives a claim and evidence, with partial reasoning.                                                                                       | Gives a claim with little evidence or reasoning.                                                             |

|                                                |                                                                                                                                                         |                                                                          |                                                                    |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------|
| <p> Safer, smarter choices and reflection </p> | <p> Gives a realistic example of a safe choice grounded in Chemistry and thoughtfully compares current understanding with the Lesson 1 prediction. </p> | <p> Gives a safe choice example and a basic reflection on learning. </p> | <p> Gives no clear example of a safe choice or no reflection. </p> |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------|

### ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

**Licensing & Copyright:** © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.